# Assignment - 7

① The intensity of sunlight hitting the Earth is about 1300 w/m². If sunlight strikes a perfect absorber, what pressure does it exert? What about a perfect reflector? What fract^n of atmospheric pressure does this amount to?

Ans: Perfect Absorber:

$$P_a = \frac{I}{c} = \frac{1300}{3 \times 10^8} = \boxed{4.33 \times 10^{-6} \text{ Pa}}$$ (Ans)

Perfect Reflector:

$$P_r = \frac{2I}{c} = \boxed{8.67 \times 10^{-6} \text{ Pa}}$$ (Ans)

Fract^n of atmospheric pressure: [$P_{atm} \approx 1.01 \times 10^{-5}$ Pa]

Absorber: $$f_a = \frac{P_a}{P_{atm}} = \frac{4.33 \times 10^{-6}}{1.01 \times 10^5} \approx \boxed{4.33 \times 10^{-11}}$$ (Ans)

Reflector: $$f_r = \frac{P_r}{P_{atm}} = \boxed{8.67 \times 10^{-11}}$$ (Ans)

② The sunlight in $\boxed{Q1}$ maybe approximated as plane-waves given by:

$$\bar{E} = E_0 \cos(\omega t - kz)\,\hat{x} \quad \& \quad \bar{B} = \left(\frac{E_0}{c}\right) \cos(\omega t - kz)\,\hat{y}$$

(a) Find the amplitude $E_0$ of the electric field

(b) Write the force experienced by an $e^-$, initially @ rest.

(c) For non-relativistic motion, find the velocity $v(t)$ of

(d) At what incident intensity, the $e^-$'s max. speed the $e^-$

would be 0.01 c ?

Ans : (a) $I = \frac{1}{2} c \epsilon_0 E_0^2$

$\Rightarrow E_0^2 = \left(\frac{2P}{c}\right) \cdot \frac{1}{\epsilon_0}$

$= \frac{8.67 \times 10^{-6}}{8.85 \times 10^{-12}}$

$\Rightarrow \boxed{E_0 \approx 10^3 \ N/C}$ (Ans)

(b) $F = q(\bar{E} + \bar{v} \times \bar{B})$

The $e^-$ is initially @ rest, hence $v = 0$ then when the $e^-$ starts moving, for $v << c$,

$|\bar{v} \times B| << |\bar{E}|$ as $B_0 = \frac{E_0}{c}$. Hence, $\bar{v} \times \bar{B}$ can be ignored

$\Rightarrow vB_0 << E_0$ for non-relativistic speeds.

$\therefore \boxed{\bar{F} = -eE_0 \cos(\omega t - kz)\,\hat{x}}$ (Ans)

(c) $\bar{F} = qE_0 \cos(\omega t - kz)\,\hat{x}$

The force is in the x-direction, hence the $e^-$ moves only in the x-direct$^n$. Since $\bar{v} \times \bar{B} \approx 0$, $(\hat{x} \times \hat{y}) = \hat{z}$ no movement in z-direct$^n$.

Hence $z \approx 0$,

$\therefore F = qE_0 \cos \omega t$

$\Rightarrow m\frac{dv}{dt} = -eE_0 \cos \omega t$

$\Rightarrow \int_0^v dv = -\frac{eE_0}{m}\int_0^t \cos \omega t \, dt$

$\Rightarrow \boxed{v(t) = -\frac{eE_0}{m\omega} \sin \omega t}$ (Ans)

(d) $v_{max} = \frac{eE_0}{m\omega} = 10^{+6} \times 3 = 0.01c$

$\Rightarrow E_0 = v_{max} \times \frac{m\omega}{e} = \frac{3 \times 10^6 \times 9.1 \times 10^{-31} \times 3 \times 10^{15}}{1.6 \times 10^{-19}}$

($\omega \approx 3 \times 10^{15}$ rad/s for visible light)

$\therefore \Rightarrow \quad B_0 \approx 5.1 \times 10^{10}$ N/mC

$\therefore \quad I = \frac{1}{2} c \epsilon_0 B_0^2$

$= \frac{1}{2} \times 3 \times 10^8 \times 8.85 \times 10^{-12} \times 5.1 \times 5.1 \times 10^{20}$

$= 10^{16} \times 345.28$

$$\boxed{I \approx 3.45 \times 10^{18} \text{ W/m}^2}$$ (Ans)

This is much higher than the intensity imparted by sunlight.